Where Is the Speaker? Silence-Informed Localization, Distance Cues and Selective Beamforming on a 4-Mic ReSpeaker

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Abstract

Service robots often hear through a small circular microphone array but use it as a single microphone. We study what the four raw microphones of a ReSpeaker USB v2.0 (6.4 cm aperture) can add to a robot's speech pipeline: the talker's angle, the talker's distance, and the suppression of sound from other directions before neural denoising. The common thread is a *silence model*: the pitch-based voice activity detector the pipeline already runs splits the stream into voice and silence, and the silence frames give the cross-spectra of everything that is not the talker. In image-source simulations of a reverberant room with a voiced talker, using that model twice in SRP-PHAT – subtracting the noise cross-spectra and weighting bins by speech presence – reduces the angle error at 0 dB SNR with a directional fan from 68.6/75.7° (plain SRP-PHAT, RT_{60} 0.3/0.6 s) to 4.2/2.6°, and a streaming tracker follows a moving talker with a median error of 1.0°. Distance cannot come from time differences on this aperture; the coherent-to-diffuse ratio is the useful cue (Spearman -0.77 with log-distance across rooms), but it is room-relative: cross-room distance error is 0.55 m with CDR and level and 0.33 m once the room's RT60 is given as a covariate, against 0.75 m for predicting the mean. An MVDR beamformer steered at the estimated talker, with its noise covariance learned while the target is silent, improves output SINR against a competing talker by 7.4 dB on average (2.6–10.8), while delay-and-sum gains 0.2 dB. All results are simulation-only; the code ships a live radar, recording and calibration tools for the hardware evaluation, and a plan to integrate the method into a ROS 2 HRI pipeline.

1 Introduction

Robots that take spoken commands in homes and competitions face the cocktail-party problem with very little hardware. A common choice is the ReSpeaker USB Mic Array v2.0: four omnidirectional microphones on a 32 mm-radius circle with an on-chip DSP that outputs one processed channel and a firmware direction-of-arrival (DOA) estimate. Our robot's speech pipeline uses exactly that processed channel: it feeds a neural denoiser (DeepFilterNet3 [9]), then a voice activity detector (VAD) built from an energy gate, an autocorrelation pitch test in 85–255 Hz and an MFCC speaker lock, then speech recognition. Spatial information is discarded, and the published firmware DOA is smoothed with a linear average, which maps a talker crossing 0° to 180°.

We ask what the raw microphones can add, in three steps a robot needs: *where* the talker is (angle), *how far* (distance), and *attending* to them by suppressing other directions before the denoiser. Our organising idea is that the VAD already provides the most useful reference signal: frames without voice describe the room – fans, motors, diffuse noise – and their cross-spectra can be used to clean the localisation and to train the spatial filter.

Contributions: (i) a streaming localiser (pitch VAD → silence model → SRP-PHAT → distance features → wrap-safe tracker) with an ablation of each use of the silence model; (ii) an analysis of which acoustic cues give distance on a 6.4 cm array, showing that the coherent-to-diffuse ratio works within a room and that a room covariate (RT60) is what makes it transfer; (iii) a measurement of MVDR versus delay-and-sum selectivity on this aperture with an estimated, not oracle, steering direction; (iv) tools for the hardware study (live radar, labelled recording, calibration) and an integration plan for a ROS 2 pipeline. All quantitative results here are from simulation; the hardware evaluation is the next step and the paper says where we expect simulation to be optimistic.

2 Related work

DOA with small arrays. GCC-PHAT [6] and steered-response power with PHAT weighting (SRP-PHAT) [3] are the standard robust estimators under reverberation; ODAS [4] runs SRP-PHAT with tracking on the same ReSpeaker and similar arrays for robots. Speech-presence or SNR-dependent weighting of time–frequency bins is a known improvement for GCC-type methods in noise; we evaluate it, and noise cross-spectrum subtraction, as two uses of one silence model.

Distance estimation. Source distance from a compact array relies on the direct-to-reverberant ratio (DRR), level, or learned features [11]. The coherent-to-diffuse ratio (CDR) is a coherence-based DRR proxy with closed-form estimators [10]; it depends on the room's critical distance, which scales with $\sqrt{V/\mathrm{RT}_{60}}$ [7].

Spatial filtering before enhancement. MVDR beamforming [2] followed by single-channel enhancement is the classic multichannel speech-enhancement chain; small-aperture arrays need superdirective designs and are sensitive to steering errors. Neural single-channel denoisers such as DeepFilterNet3 [9] remove stationary and diffuse noise but cannot separate two talkers by direction.

Simulation. We use the image-source method [1] via pyroomacoustics [8], and a diffuse noise field with the exact spherical-isotropic coherence [5].

3 Method

Signal model. Four microphones p_m on a 32 mm circle, 16 kHz, 512-point Hann STFT, hop 256. A far-field source at azimuth θ gives $X_m(f) = S(f)e^{-j2\pi f\tau_m(\theta)}$, $\tau_m = -p_m^\top u(\theta)/c$. The largest inter-microphone delay is 0.19 ms (≈3 samples).

VAD and silence model. Per frame, the mean of the four channels over the last 1024 samples passes an energy gate and an autocorrelation test: the normalised autocorrelation peak in the lag band of 85–255 Hz must exceed 0.6 (the robot pipeline's rule), which also yields F_0. The gate is $\max(g_0, \phi \cdot 10^{6/20})$ where ϕ follows the RMS of non-speech frames (fast down, slow up), because the raw microphones are not gain-controlled. Non-speech frames, after a 0.3 s hangover, update the noise cross-spectral matrix Φ_n (time constant 2 s). Speech frames update Φ_x^{fast} (0.25 s) and Φ_x^{slow} (1 s); both are reset when speech resumes after 0.6 s of silence.

Angle. SRP-PHAT on a 1° grid in 200–3800 Hz:

$$P(\theta) \propto \sum_{i<j} \sum_f w_f \,\mathrm{Re}\Big\{ \tfrac{R_{ij}(f)}{|R_{ij}(f)|} e^{j2\pi f(\tau_i(\theta)-\tau_j(\theta))} \Big\},$$

with $R = \Phi_x^{\text{fast}} - \Phi_n$ (noise subtraction) and $w_f = \max(0, 1 - \mathrm{tr}\,\Phi_n/\mathrm{tr}\,\Phi_x)$ (speech-presence weight). Optional: a Gaussian comb at multiples of F_0. The peak is refined parabolically; confidence is the peak's height above the map mean. Baseline: per-pair GCC-PHAT upsampled 16× and a least-squares fit of $u(\theta)$.

Distance. From $\Phi_s = \Phi_x^{\text{slow}} - \Phi_n$: (a) CDR with the DOA-independent estimator of [10] per pair, using bins where the diffuse coherence $\mathrm{sinc}(2\pi f d/c) < 0.7$ in 0.4–7 kHz, median over bins and over the six pairs; (b) speech level (dBFS); (c) SNR over Φ_n; (d) spectral tilt. A ridge regression maps standardised features to $\log r$; RT_{60} can be appended as a room covariate (the ReSpeaker firmware reports an estimate).

DOA	8 angles × SNR {0,10,20} × RT_{60} {0.3,0.6} × {diffuse, fan}
Distance	6 distances 0.5–3 m × 3 angles × RT_{60} {0.3,0.45,0.6} × 2
Beam	interferer at 45–180°, SIR 0 dB, RT_{60} {0.3,0.6}
Demo	7 positions, 0.5–3 m, RT_{60} 0.45, 17.5 s

Table 1: Local simulation sweeps (`local_small.yaml`).

Tracking. An alpha-beta filter on the circle (residuals wrapped to $[-180°, 180°)$) and an EMA of $\log r$; measurements below a confidence threshold are ignored, and tracks expire after 3 s.

Selective filtering. MVDR $w = (\Phi_n + \delta I)^{-1}a/(a^H(\Phi_n + \delta I)^{-1}a)$ steered at the estimated azimuth, $\delta = 1\%$ of per-bin power, against delay-and-sum $w = a/4$. In the beamforming experiment the target's azimuth is estimated by SRP-PHAT on the difference between cross-spectra after and before the target's onset, and Φ_n is the pre-onset covariance (interferer plus noise).

Simulation. A shoebox room (8×7×3 m, image-source order ≤25, Sabine absorption for the target RT_{60}), array at 1.1 m height placed off-centre, mouth at 1.5 m. The talker is a synthetic voiced signal (random F_0 in 100–220 Hz, vowel formants, syllabic envelope, pauses), −30 dBFS at 1 m with ±4 dB effort jitter in the distance sweep. Noise: spherically diffuse pink noise [5] plus mic self-noise, optionally a directional pink "fan" at 200°, 2 m. Sweeps are listed in Table 1.

4 Results

Angle. Figure 1. At 20 dB SNR all estimators are within about 2°. At 0 dB the two uses of the silence model matter in different conditions. With diffuse noise, dropping the speech-presence weights raises the error from 3.2/3.0° (RT_{60} 0.3/0.6 s) to 19.4/26.5°: after subtraction, noise-only bins carry structured residue that PHAT normalises to full weight. With the fan, dropping the subtraction raises it from 4.2/2.6° to 7.8/20.1°, and plain SRP-PHAT locks on the fan (68.6/75.7°). GCC-PHAT with least squares reaches 21.7–34.3° at 0 dB. The F_0-harmonic comb changes nothing measurable on our synthetic talker, whose spectrum is already purely harmonic.

Tracking a moving talker. Figure 2: after 0.5 s of settling per position, the tracked azimuth has a median

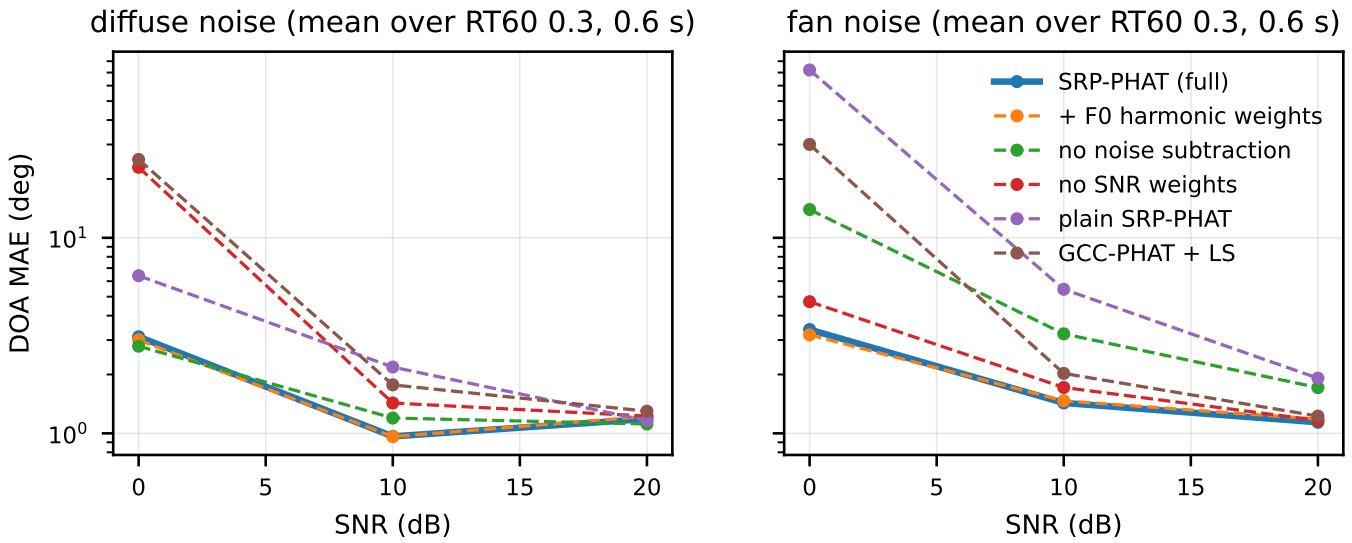


Figure 1: DOA mean absolute error (log scale) versus SNR, averaged over RT_{60} 0.3 and 0.6 s, talker at 1.5 m. Removing either use of the silence model hurts in one noise condition; removing both fails with a directional noise source.

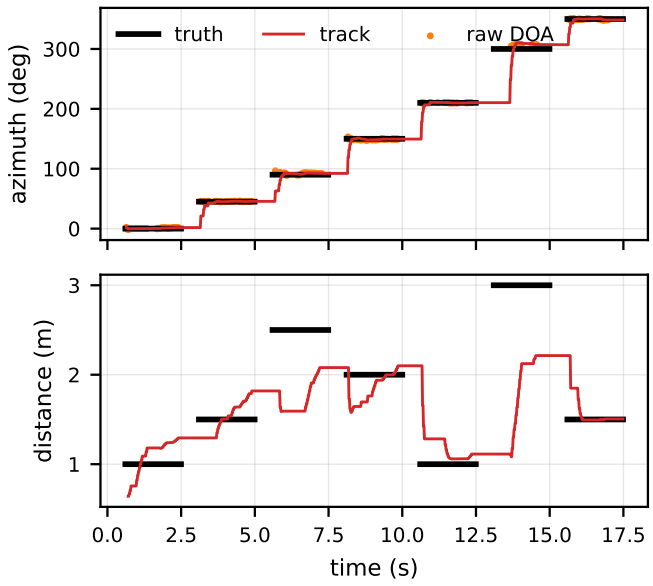


Figure 2: Streaming estimates for a talker moving through 7 positions (black: truth). Top: azimuth. Bottom: distance.

error of 1.0° (90th percentile 7.0°). The distance track (model trained in the same room) has a median error of 0.18 m and under-estimates the 3 m position.

Distance. Figure 3: the CDR is the strongest cue ($\rho = -0.77$ pooled, -0.90 to -0.95 within a room). Level ($\rho = -0.61$) suffers from the ±4 dB vocal-effort jitter, SNR (−0.48) from how much speech leaks into the noise estimate, and tilt (+0.13) carries little information. Figure 4: across angles, CDR alone gives 0.44 m MAE and CDR + level 0.45 m, against 0.75 m for predicting the mean and 0.90 m for an inverse-square law on level. Adding the room's RT_{60} gives 0.24 m, and across rooms (leave one RT_{60} out) 0.33 m versus 0.55 m without it.

Selective cancellation. Figure 5: MVDR gains 2.6–10.8 dB (mean 7.4 dB), less in the more reverberant room and at 45° separation, where the estimated target DOA was also furthest off (9.6°). Delay-and-sum gains 0.2 dB: at 1 kHz the array spans 0.19 wavelengths, so only superdirective weights give selectivity.

5 Discussion

The silence model is the method. The estimators are textbook. What made them work on this array was getting "noise" right: which bins contain it, which frames it can be learned from (a fixed energy gate tuned for the processed channel missed a talker at 3 m entirely), and when old statistics stop describing the scene (a talker who moved between utterances). The robot pipeline's VAD already provides the voice/silence split, so these gains come without new sensing.

Distance is a room-calibrated quantity. A 6.4 cm array has no geometric range resolution; the CDR measures distance relative to the room's critical distance. This predicts, and the cross-room experiment confirms, that a room descriptor is needed. The ReSpeaker firmware's RT_{60} estimate is a candidate; its accuracy is untested. Without it, a short calibration per room (a few labelled utterances) or coarse near/mid/far buckets are the realistic outputs.

Where simulation is optimistic. The synthetic talker is more periodic and spectrally simpler than speech, which helps both the VAD and PHAT; the shoebox image-source model has specular reflections and no furniture, scattering or robot body; the diffuse field is ideal; and the microphones are perfectly matched and positioned.

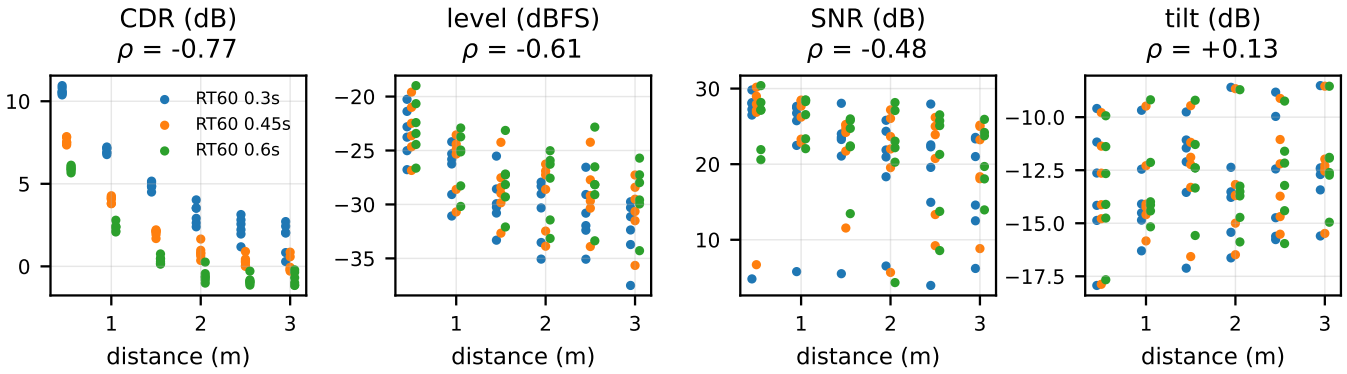


Figure 3: Distance features versus true distance (108 clips), coloured by RT_{60}; ρ is Spearman's correlation with $\log r$ over all rooms. CDR is monotone within each room but offset between rooms.

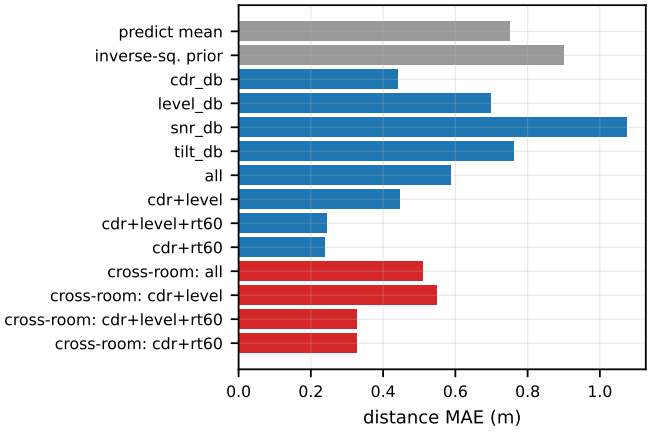


Figure 4: Distance MAE. Blue: leave-one-angle-out; red: leave-one-RT_{60}-out.

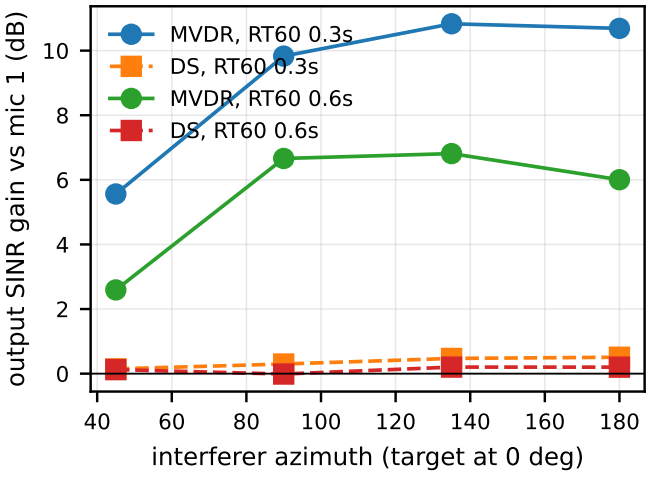


Figure 5: Output SINR gain over microphone 1 against a competing talker (target at 0°, both at 1.2 m, SIR 0 dB), with steering from the estimated DOA.

Real recordings will also reveal the mapping between raw channels, our azimuth convention and the firmware's. We expect larger angle errors at low SNR and a weaker within-room CDR correlation on hardware.

Limitations. Small sweeps (8 clips per DOA cell, 108 distance clips, 8 beamforming scenes) without confidence intervals; a larger configuration is provided. One interferer, equal levels; no moving interferer; no own-voice (TTS) handling; MVDR's noise covariance in the beamforming experiment comes from a clean pre-onset segment.

6 Conclusion

On a four-microphone, 6.4 cm array, the voice/silence split a robot's speech pipeline already computes is enough to turn the raw channels into a usable sense of where the talker is: a few degrees of angle error at 0 dB SNR when the silence frames are used both to subtract noise and to weight frequency bins, distance within about a third of a metre once the room's reverberation is known, and several dB of suppression of a competing talker with an MVDR filter steered at the estimate. Next steps are the hardware evaluation with labelled recordings (including the firmware DOA as a baseline and its RT_{60} estimate as the room covariate), and integrating the localiser and spatial filter as ROS 2 nodes ahead of the existing DeepFilterNet3 denoiser, evaluated by speech recognition accuracy with two simultaneous talkers.

Reproducibility. `make all` in `projects/15_spatial_audio_attention` regenerates every number and figure from the simulation in about three minutes on a laptop CPU; `make live`, `make record` and `make calibrate` run the hardware study.

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